

Mechanism-driven drug repurposing with integrated regulatory, prevalence and clinical-network evidence recovers known indications and proposes actionable combinations

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Abstract

Computational drug-repurposing screens routinely propose plausible drug-disease pairings, yet most candidates fail at the next layer of scrutiny: the indication is already on-label, the receptor is absent from the tissue, the patent has expired, no clinician is positioned to lead a trial. We built an end-to-end screen that scores 3,996 approved drugs against 28,198 diseases (175,759 substance-disease hypotheses after active-ingredient grouping) and layers on directionality, tissue expression, orthologous-gene model-organism evidence, Reactome pathway co-membership, FDA Orange Book intellectual-property runway, Orphanet rare-disease prevalence, PubMed/Europe PMC/ClinicalTrials.gov literature priors, regional Key Opinion Leader identification, and an automated combination-therapy companion finder. Validation against 19 curated known repurposing successes recovered 16 (84%) at mechanism-support threshold 0.3; the three misses traced to specific Open Targets coverage gaps for rare or peripheral indications. The screen surfaced the BRAF + MEK doublet *de novo* in cardio-faciocutaneous syndrome — the same combination already standard-of-care in BRAF V600E metastatic melanoma — and identified metformin’s metabolic effect on polycystic ovary syndrome via mitochondrial complex I (NDUFA13, MT-ND4) rather than the AMPK route the literature most often invokes, recapitulating the proximal pharmacological mechanism. Severity-aware damping correctly eliminated 32 receptor-LoF + agonist false positives (the insulin → INSR-deficiency cluster) while preserving partial-LoF rescuable cases (TZD → PPARG-related lipodystrophy). End-to-end runtime is approximately 12 minutes on a laptop; output PDFs identify a regional KOL and propose a tractable collaboration structure per hypothesis.

Introduction

Repurposing — administering an approved drug to a new indication — has produced some of the most consequential pharmacological advances of the past four decades: sildenafil from angina to pulmonary arterial hypertension¹, thalidomide from morning sickness to multiple myeloma², propranolol from hypertension to infantile haemangioma³, methotrexate from oncology to rheumatoid arthritis⁴. The economic and clinical case for systematic repurposing is now well-established^{5,6}: an approved drug carries through safety and ADME validation, shortens development by several years, and reduces capital-at-risk by roughly an order of magnitude relative to *de novo* discovery.

In-silico screens have proliferated alongside the public biomedical resources that enable them. PREDICT used drug-drug and disease-disease similarity matrices to suggest novel pairings⁷. DrugCentral aggregates regulatory and mechanism curation⁸. The Connectivity Map and L1000 expression-signature platform exploit transcriptional perturbation profiles⁹. More recent graph-neural-network and network-pharmacology approaches integrate heterogeneous evidence at scale^{10,11}.

Open Targets has become the *de facto* aggregator for target-disease association^{12,13}, and ChEMBL the canonical source of mechanism-of-action annotation across the approved-drug universe^{14,15}.

A persistent weakness across this family of tools, however, is that they end at a mechanism score. Useful as that is, a score alone cannot answer the questions that determine whether a candidate is actionable: Is the indication actually novel? Does the drug push the target in a direction the disease’s genetic mechanism predicts is therapeutic? Is the target expressed in the disease tissue? For monogenic disorders, can the drug engage what the disease has broken? What is the substance’s IP runway, and is there generic competition? What is the patient population — and does it qualify for orphan exclusivity? Has the literature already saturated this hypothesis? Which clinician is the natural lead investigator? These are the questions a fund partner or clinical-development lead asks before a repurposing programme advances.

Here we describe a screen designed to answer all of these in a single pipeline. We use only free public data, validate the mechanism layer against 19 known repurposing successes, and ship the candidate per-hypothesis as a PDF brief identifying a regional Key Opinion Leader and proposing a tractable collaboration model. The screen scales to all of approved pharma × all of disease ontology in ~12 minutes on a laptop.

Results

Full-screen output

A single end-to-end run on Open Targets release 24.06 produced 200,000 ranked candidate (drug, disease) pairs (capped from a larger search space at minimum opportunity 0.05), reduced by active-ingredient grouping to 175,759 substance-disease rows. Substance grouping used the ChEMBL `molecule_hierarchy` table to collapse formulations such as PIOGLITAZONE HYDROCHLORIDE and PIOGLITAZONE up to their parent active ingredient, with variant names listed in a side column. Of these, 128,313 received a Reactome pathway co-membership boost (only indirect bridges — where the drug’s target differed from the disease’s strongly associated target — were credited, to avoid double-counting the direct-target signal); 13,740 received an orthologous-gene phylogenetic boost from the IMPC/PhenoDigm data^{16,17}; 32,292 carried market-size data (curated US sources plus Orphanet¹⁸); 62,210 carried FDA Orange Book IP signals (NA for biologics, correctly so — they live in the Purple Book); and 32 were damped by a “receptor LoF + agonist” severity heuristic (Section *Severity-aware damping*, below).

Backtest validation: 16 of 19 known repurposing successes recovered

To measure whether the screen’s mechanism layer actually surfaces real repurposing connections, we curated 19 known successes (Table 1) and computed `mech_support` — the noisy-OR over drug-target × target-disease contributions¹⁹ — exactly as the screen does, but without subtracting novelty (since the repurposed indications are now on-label and would otherwise be filtered). At a threshold of 0.3, sixteen of nineteen cases scored HIT (Table 1, **84% recovery rate**), including sildenafil → pulmonary arterial hypertension (0.61, via PDE5A), thalidomide → multiple myeloma (0.98, via CRBN), methotrexate → rheumatoid arthritis (0.85), tofacitinib → ulcerative colitis (0.98, via JAK1/2/3), and naltrexone → alcohol dependence (1.00, via OPRM1/OPRK1).

The three failures all traced to Open Targets data-coverage gaps: minoxidil → alopecia (K-ATP follicle effect not curated as a disease association), propranolol → capillary infantile haemangioma (rare condition with weak target-disease signal), and acetazolamide → idiopathic intracranial hypertension (disease absent from OT entirely). The failures are not pipeline-logic errors but resource-coverage limits, which lift automatically with each OT release.

The validation script flagged one case as a “surprising hit”: metformin → polycystic ovary syndrome scored 1.00, but the contributing targets were mitochondrial respiratory-complex-I genes (MT-ND2, MT-ND4, NDUFA13, NDUFB1) rather than the AMPK subunits PRKAA1/PRKAA2 the literature most often invokes. Metformin’s primary pharmacological action is in fact complex-I inhibition^{20,21}, with AMPK activation a downstream consequence — the screen surfaced the proximal mechanism, not the convenient downstream one.

***De novo* discovery of established combination therapies in adjacent indications**

The combination-therapy companion finder evaluates, for each top-N primary hypothesis, candidate companion substances whose direct targets bridge strong disease targets the primary does not hit. Synergy is measured as `combo_mech - mech_primary` (the noisy-OR over the union of contributions, minus the primary's coverage alone), with a target-overlap filter to reject same-class redundancy.

Strikingly, the screen rediscovered the BRAF + MEK inhibitor doublet — currently standard-of-care for BRAF V600E metastatic melanoma^{22,23} — from first principles in cardiofaciocutaneous syndrome, a developmental RASopathy driven by germline pathway-activating mutations in BRAF, MAP2K1, KRAS, or NRAS²⁴. All three BRAF inhibitors in the approved-drug set (dabrafenib, vemurafenib, encorafenib) paired with binimetinib (and to a lesser extent cobimetinib) at synergy 0.15; the KRAS analogues sotorasib and adagrasib paired with binimetinib at synergy 0.17 (Table 2). Treating CFC syndrome (~1 in 270,000 births, an orphan indication) with the pathway-targeting agents already in clinical use for melanoma is a research-active hypothesis²⁵, and the screen identified it without prior programming of any oncology-developmental cross-reference.

Other notable combinations the screen surfaced (Table 2): - **Bromazepam + orphenadrine → developmental and epileptic encephalopathy** (benzodiazepine GABA-A potentiation + NMDA antagonism, synergy 0.48 — the highest in the top 30); - **Angiotensin II + aliskiren → renal tubular dysgenesis** (RAS axis combination, bridge target REN, synergy 0.21); - **Pinacidil/minoxidil + vernakalant → familial atrial fibrillation** (K-ATP opener + atrial-specific K⁺ blocker); - **Setmelanotide/bremelanotide + metformin → type 2 diabetes mellitus** (MC4R agonist + complex-I inhibitor; bridge target NDUFB1 — the same proximal mechanism the metformin/PCOS backtest case revealed).

Of 200 primary hypotheses in the combination-finder pass, 61 received at least one companion satisfying the synergy threshold and target-overlap filter.

Severity-aware damping eliminates the “receptor LoF + agonist” trap

Direct-target mechanism scoring produces a recurring failure mode in monogenic disorders: when the disease is caused by loss-of-function of a receptor, an agonist for that receptor cannot rescue the broken protein. Without correction, an early version of the screen had the insulin formulations clustered at ranks 11–30 of the full output paired with “hyperinsulinism due to INSR deficiency”, and G-CSF analogues paired with “autosomal recessive severe congenital neutropenia due to CSF3R deficiency” — both biologically futile.

We implemented a narrow, conservative heuristic: when a gene named in the disease is also a direct target of the drug (`disease_gene_match`), the disease name contains severity language (“deficiency”, “complete absence”, “Donohue syndrome”, “Rabson-Mendenhall”²⁶), AND the drug's action class is agonistic, the hypothesis is flagged `severe_loF_agonist` and its opportunity is reduced by 70%. The cluster (32 hypotheses) was pushed below rank 100 — effectively out of the screen — while adjacent partial-LoF rescuable cases were preserved: thiazolidinediones for PPARG-related familial partial lipodystrophy (FPLD3), where TZD agonism rescues residual receptor activity²⁷, remained at rank 30; calcimimetics for familial hypocalciuric hypercalcemia 1, where partial CaSR function persists²⁸, remained at ranks 19–20. The discrimination rests entirely on the disease name's severity language — INSR-deficiency hyperinsulinism implies a severe-LoF state, “partial lipodystrophy” or “hypocalciuric” does not.

Filtering noisy direction signal from model-organism sources

A second class of failure mode arose from how Open Targets aggregates direction-of-effect evidence. Of approximately 1.02 million direction-informative rows in the OT evidence parquet, ~1.14 million originate from IMPC (mouse-knockout phenotyping via PhenoDigm), versus only ~166,000 from the human-genetics direction sources (`ot_genetics_portal`) and ~282,000 from ClinVar (eva). IMPC's `variantEffect = LoF / directionOnTrait = risk` calls — appropriate for “what happens when you delete the mouse gene” — frequently flip the inferred therapeutic direction for human disorders whose mechanism differs from a complete knockout. For EPOR + primary familial polycythemia (a *gain*-of-function truncation

that removes the receptor’s negative-regulatory domain²⁹), IMPC contributed 16 LoF/risk rows; the aggregated direction signal recommended an EPOR *agonist* — the opposite of the correct therapy. After filtering the direction adapter to seven curated human-genetics sources (ot_genetics_portal, gene_burden, eva, gene2phenotype, clingen, genomics_england, orphanet), the EPO → polycythemia cluster fell 275 ranks (22–30 → 297–301), and the direction CSV more broadly contracted from ~1.02M rows to 113,339 with a balanced direction split (64,294 +1 vs 49,045 –1, versus the previous ~95% skew toward +1).

Operational outputs

The screen produces a 36-column CSV that is hard for non-technical stakeholders to act on directly. We therefore added a one-page A4 PDF generator that, per hypothesis, weaves the mechanism evidence into prose (“Drug-target convergence on the disease scores 0.93...”), translates the Orange Book columns into IP-runway commentary (“LOE 2037 with no generic competition — meaningful IP runway for a co-development conversation”), expresses market opportunity with prevalence-source attribution (“approximately 16,650 US patients — below the 200,000 Orphan Drug Act threshold”), identifies a regional Key Opinion Leader from PubMed authorship with h-index from Semantic Scholar³⁰ and contact email when present in the affiliation, and proposes a three-way collaboration structure (KOL → bioinformatics analytics partner → IP-holder substance owner, under an investigator-initiated trial framework). For the Bromazepam → developmental and epileptic encephalopathy hit, the screen identified Boris Keren (Département de Génétique, Pitié-Salpêtrière, Paris; h-index 56) as the EU KOL — a real specialist whose publication record on developmental-epilepsy genetics aligns with the hypothesis’s mechanism. Regional preference was tuned to favour Boston/Cambridge-MA/Bay Area in the US and Belgium > Netherlands > France/Germany/UK in the EU.

Discussion

The technical content of each enrichment layer in isolation is not novel: novelty subtraction via ontology hops, directionality inference from genetic evidence, tissue-expression gating, pathway-overlap analysis, and PubMed-based investigation priors are all individually established. The contribution of this work is the *integration* — combining these layers into a single pipeline with measured precision (84% recovery of known repurposing wins), with the regulatory and economic signals (IP, prevalence, generic status) that determine whether a candidate is actionable, with the human-network signal (identified KOLs) that determines whether it is approachable, and with the operational artefact (PDF deal briefs) that determines whether the work leaves the screen.

The validation framework is also worth emphasising. Many repurposing screens are presented with anecdotal example hits; here we measure precision against an explicit ground-truth set, characterise the failure modes (all three misses are OT data-coverage gaps), and ship the measurement script alongside the pipeline so any user can re-verify or extend. The metformin/PCOS surprising-hit case is, in our view, a small piece of evidence that the screen is doing genuine mechanistic inference rather than retrieval — it surfaces the proximal pharmacological mechanism rather than the post-hoc literature attribution.

The combination-therapy result deserves note. Re-deriving the BRAF + MEK doublet for cardiofaciocutaneous syndrome from first principles — without any oncology-developmental cross-reference programmed in — is a particularly clean form of internal validation. The screen’s noisy-OR formulation across pathway-bridging companion targets recovers an established clinical strategy in an adjacent indication.

Limitations are real. Open Targets’ disease-association coverage is the precision ceiling, and our three backtest misses correspond exactly to under-curated rare or peripheral indications. The severity heuristic relies on disease-name keywords rather than structured mechanism-of-disease annotations from, e.g., ClinGen; expanding to structured severity will require curated input. KOL disambiguation depends on Semantic Scholar’s author records, which use abbreviated first names and can collide on common surnames — we resolve this by selecting the highest h-index among matches, but the heuristic cannot recover a true KOL who is genuinely junior. The literature pass at full coverage (>100,000 hypotheses) would

require either a multi-day cache build or a more parallel scheduler than the current rate-limiter permits. The Orange Book is small-molecule only; the FDA Purple Book equivalent for biologics is an obvious extension.

The pipeline is engineered for incremental addition. Natural next layers include cell-line / DepMap evidence for cancer hypotheses³¹, structured drug-drug interaction filtering for combination-therapy safety³², multi-omics-based patient-stratification analyses for the operational output, and a continuously-updated web dashboard tied to a nightly refresh of the upstream sources.

The full pipeline source, configuration, curated CSVs, validation YAML, and this manuscript are publicly available at <https://github.ugent.be/wvcrieki/repurposed>.

Methods (condensed)

Data sources. ChEMBL 37¹⁴ (approved drugs, mechanism of action, indications, molecule_hierarchy), Open Targets 24.06¹² (target-disease associations, evidence, diseases, baseline expression, targets parquet with integrated Reactome³³ pathways), STRING v12.0³⁴ (protein-protein interaction edges), EFO 2024-06³⁵ (disease ontology), FDA Orange Book (May 2026 release), Orphanet (en_product9_prev.xml, CC-BY-4.0)¹⁸, PubMed E-utilities, Europe PMC REST, ClinicalTrials.gov v2 API, Lens.org Patent API (optional), Semantic Scholar Author API³⁰.

Mechanistic scoring. Per (drug, disease), $\text{mech_support} = 1 - \prod_i (1 - \text{contrib}_i)$ (noisy-OR¹⁹) where each contribution is $\text{target_weight} \times \text{assoc_score}$ over the drug's direct targets (weight 1.0) optionally augmented by STRING neighbours (weight $0.5 \times \text{STRING_score}$). Open Targets assoc_score filtered at ≥ 0.1 .

Novelty. Ontology distance from the drug's nearest known indication via EFO graph³⁵ walked outward to radius 1. ChEMBL drug indications normalised colon→underscore (MONDO:0005148 → MONDO_0005148) to align with OT format. Target-class indication rollup: drug A inherits drug B's indications iff A's frozenset of (direct target, action_type) pairs is a subset of B's; STRING neighbours excluded from class keys.

Direction. Filtered Open Targets evidence to ot_genetics_portal, gene_burden, eva, gene2phenotype, clingen, genomics_england, orphanet only (IMPC and other non-human-genetics sources excluded). Per (target, disease), $\text{variant_Effect} \in \{\text{LoF}, \text{GoF}\}$ combined with $\text{directionOnTrait} \in \{\text{risk}, \text{protective}\}$ gives therapeutic direction in $\{+1, -1\}$ per the OT documented inference¹².

Tissue. Per (target, disease), score is max over the disease's relevant tissues of $(\text{disease_relevance} \times \text{target_expression})$. Bucketed into expressed (≥ 0.25), low (linear ramp), absent (measured zero, factor 0.3), unknown (no measurement, factor 0.7). Multiplicative.

Phylogenetic boost. From OT evidence datatypeId == animal_model (default IMPC¹⁶), per (drug, disease) the max phylo_score across the drug's targets. $\text{phylo_factor} = 1 + 0.5 \times \text{phylo_score}$. Asymmetric: presence boosts, absence does not penalise.

Pathway boost. Reactome³³ pathway co-membership per (drug, disease), restricted to pathways with ≤ 80 genes (specificity filter). Only drug_target $\neq$ disease_target bridges credited ($\text{indirect_only} = \text{true}$, to avoid double-credit of direct hits). $\text{pathway_factor} = 1 + 0.3 \times \log_{10}(\text{n_overlap}) / \log_{10}(5)$.

Opportunity score. $\text{mech}^1 \times \text{novelty}^1 \times \text{direction} \times \text{tissue} \times \text{phylo} \times \text{pathway} / \sqrt{\text{n_drug_targets}}$.

Substance grouping. ChEMBL molecule_hierarchy.parent_molregno rolls formulations up to active ingredient.

Severity damping. When disease_gene_match populated AND disease name contains severity language AND drug action is agonistic, $\text{opportunity} \times 0.3$, re-rank.

Literature pass. Per top-N pair (default 5,000), query PubMed/Europe PMC/ClinicalTrials.gov/Lens with synonym-expanded queries. Counts log-saturated and combined to $\text{investigation_prior} \in [0, 1]$. Opportunity damped by $(1 - 0.5 \times \text{investigation_prior})$.

Market sizing. Curated CSV (122 major US diseases from CDC/SEER/foundation estimates) overlaid by Orphanet structured prevalence (~5,100 rare diseases bridged to OT MONDO via dbXRefs).

IP signals. FDA Orange Book parsed per active ingredient: max patent expiry, max exclusivity expiry, generic availability via any ANDA filing.

KOL identification. PubMed eSearch + efetch on top-N (target, disease) queries; affiliation regional classification (US: Boston/Cambridge-MA/Bay Area preferred; EU: Belgium > Netherlands > France/Germany/UK); h-index from Semantic Scholar³⁰ with abbreviated-name handling (E. Topol matches Eric Topol via surname + first-letter, highest h-index among matches).

Backtest validation. 19 curated cases with expected drug, repurposed indication, and mechanism targets. Per case, all ChEMBL IDs matching drug name (parent + salts) pooled; mech_support computed; HIT at ≥ 0.3 .

Output. 36-column CSV plus one-page A4 PDFs (ReportLab) per top hypothesis with identified KOL.

Compute. Full run with all enrichment layers and literature pass: approximately 12 minutes on M-series Mac (16 GB RAM, ~50 GB free disk). DuckDB engine for the streaming target-disease join³⁶.

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Author contributions

W.V.C. conceived the screen architecture, implemented the pipeline, curated the validation set, generated and analysed the results, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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